HW 2: Types of Data

CP2132 Data Models

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Import Dependency

```
import pandas as pd
import matplotlib.pyplot as plt
```

Part 1 : Basic Python

Task 1

Perform the following:

- Write a program to get input of different types from the user.
- Display type of each variable.
- Convert a variable of the type float to integer.
- Demonstrate a comparison of before vs after the type conversion.

```
# Input values from the user
string_value = input("Enter a string: ")
integer_value = int(input("Enter an integer: "))
float_value = float(input("Enter a float: "))

Enter a string: Fastjam
Enter an integer: 40
Enter a float: 76.47

# Display the data types
print("Type of string_value: {type(string_value)}")
print("Type of integer_value: {type(integer_value)}")
print("Type of float_value: {type(float_value)}")

Type of string_value: <class 'str'>
Type of integer_value: <class 'int'>
Type of float_value: <class 'float'>

# Convert a variable of the type float to integer.
original_float = float(40)
converted_integer = int(original_float)

# Demonstrate a comparison of before vs after the type conversion.
print("Original float value: {original_float} {type(original_float)}")
print("Converted integer value: {converted_integer} {type(converted_integer)}")
print("Comparison {original_float == converted_integer}: {original_float == converted_integer}")

Original float value: 76.47 {type: <class 'float'>}
Converted integer value: 76 {type: <class 'int'>}
Comparison {original_float == converted_integer}: False
```

Task 2

Given a dictionary of students with their scores, find the average score of each student across all subjects and identify the student with the highest average score.

```
students_grades = {
    "John": [85, 90, 95],
    "Bob": [70, 75, 80],
    "Mark": [75, 85, 72],
    "Diana": [90, 95, 94],
    "Charlie": [70, 65, 68]
}

# Calculate and print the average grade for each student
student_averages = {}
for student, grades in students_grades.items():
    average_grade = sum(grades) / len(grades)
    student_averages[student] = average_grade
    print(f"{student}: Average Grade = {average_grade:.2f}")

John: Average Grade = 84.33
Bob: Average Grade = 75.00
Mark: Average Grade = 77.33
Diana: Average Grade = 93.00
Charlie: Average Grade = 71.67

# Find the student with the highest average grade
highest_avg_student = None
highest_average = 0

for student, average in student_averages.items():
    if average > highest_average:
        highest_average = average
        highest_avg_student = student

print(f"Student with the highest average score: {highest_avg_student} {Average: {highest_average:.2f}}")

Student with the highest average score: Diana {Average: 93.00}
```

Part 2 : Working with CSV!

Add a new column to this CSV file named 'Bonus,' and calculate the bonus for employees in the Sales department as 10% of their MonthlyRate.

```
# Load the CSV file
file_path = 'employee_data.csv'
df = pd.read_csv(file_path)
```

```
df.head()
```

Out[10]:

	Age	DailyRate	Department	EducationField	Gender	MaritalStatus	MonthlyRate	OverTime
0	41	1102	Sales	Life Sciences	Female	Single	19479	Yes
1	49	279	Research & Development	Life Sciences	Male	Married	24987	No
2	37	1373	Research & Development	Other	Male	Single	2096	Yes
3	33	1392	Research & Development	Life Sciences	Female	Married	23159	Yes
4	27	591	Research & Development	Medical	Male	Married	16632	No

```
# Add new column call 'Bonus'
df['Bonus'] = 0.0 # Initialize Bonus column with 0

# Calculate 10% of bonus into the 'Bonus' column
df.loc[df['Department'] == 'Sales', 'Bonus'] = df['MonthlyRate'] * 0.10

# Print and show result
print("DataFrame with 'Bonus' column:")
print(df.head())

DataFrame with 'Bonus' column:
   Age  DailyRate  Department  EducationField  Gender  \
0   41       1102       Sales  Life Sciences  Female
1   49        279  Research & Development  Life Sciences    Male
2   37       1373  Research & Development           Other    Male
3   33       1392  Research & Development  Life Sciences  Female
4   27        591  Research & Development        Medical    Male

  MaritalStatus  MonthlyRate  OverTime   Bonus
0        Single        19479       Yes  1947.9
1       Married        24987        No     0.0
2        Single         2096       Yes     0.0
3       Married        23159       Yes     0.0
4       Married        16632        No     0.0

# Save the updated DataFrame back to a CSV file
output_file_path = 'employee_data_with_bonus.csv'
df.to_csv(output_file_path, index=False)
print("Updated DataFrame saved to {output_file_path}")

Updated DataFrame saved to employee_data_with_bonus.csv
```

Part 3 : Working with Matplotlib

Plot a graph showing the average salary of employees in each department to compare the average salaries across departments.

```
# Load CSV and create DataFrame
data = pd.read_csv('employee_data.csv') # Using original data for this part
df = pd.DataFrame(data)

# Calculate average MonthlyRate by Department
average_monthly_rate_by_department = df.groupby('Department')['MonthlyRate'].mean().reset_index()
print("Average MonthlyRate by Department:")
print(average_monthly_rate_by_department)

Average MonthlyRate by Department:
              Department  MonthlyRate
0  Human Resources  24635.000000
1  Research & Development  14186.865765
2                  Sales  14489.795572

# Plotting the bar chart
plt.figure(figsize=(10, 6))
plt.bar(average_monthly_rate_by_department['Department'], average_monthly_rate_by_department['MonthlyRate'], color='skyblue')
plt.xlabel('Department')
plt.ylabel('Average Monthly Rate')
plt.title('Average Monthly Rate by Department')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

Challenge!!! : Working with SATAN

Great job, for not going up on this subject!

Here's the story: The CSV file that was given to you isn't just an ordinary file – it's data from an organization where Aliya Sian works!

Aliya feels that something suspicious is going on within the company. She suspects that there's inequality in salaries between male and female employees.

She also wonders if the government might be intervening to encourage population growth by secretly increasing the salaries of married employees compared to those who are single.

Additionally, how much of a difference is there in salaries between employees who graduated from different fields of study?

Can you help Aliya prove whether her suspicions are just in her head or if they're actually true?

For Aina, everyone is truly **equal!**

```
# Department code here
# Calculate average salary by between genders
avg_salary_gender = df.groupby('Gender')['MonthlyRate'].mean().reset_index()
print("\nAverage Monthly Rate by Gender:")
print(avg_salary_gender)

# Plotting Average Monthly Rate by Gender
plt.figure(figsize=(8, 5))
import seaborn as sns
sns.barplot(x='Gender', y='MonthlyRate', data=avg_salary_gender, palette='coolwarm')
plt.title('Average Monthly Rate by Gender')
plt.xlabel('Gender')
plt.ylabel('Average Monthly Rate')
plt.show()

# Analyze salary based on MaritalStatus
avg_salary_marital = df.groupby('MaritalStatus')['MonthlyRate'].mean().reset_index()
print("\nAverage Monthly Rate by Marital Status:")
print(avg_salary_marital)

# Plotting Average Monthly Rate by Marital Status
plt.figure(figsize=(8, 5))
sns.barplot(x='MaritalStatus', y='MonthlyRate', data=avg_salary_marital, palette='viridis')
plt.title('Average Monthly Rate by Marital Status')
plt.xlabel('Marital Status')
plt.ylabel('Average Monthly Rate')
plt.show()

# Analyze salary based on EducationField
avg_salary_education_field = df.groupby('EducationField')['MonthlyRate'].mean().reset_index()
print("\nAverage Monthly Rate by Education Field")
print(avg_salary_education_field)

# Plotting Average Monthly Rate by Education Field
plt.figure(figsize=(10, 6))
sns.barplot(x='EducationField', y='MonthlyRate', data=avg_salary_education_field, palette='plasma')
plt.title('Average Monthly Rate by Education Field')
plt.xlabel('Education Field')
plt.ylabel('Average Monthly Rate')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

Average Monthly Rate by Gender:
  Gender  MonthlyRate
0 Female  14676.608340
1   Male  14872.185442

/tmp/ipykernel_1657705943.py:30: FutureWarning:

Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'x' variable to 'hue' and set 'legend=False' for the same effect.

  sns.barplot(x='Gender', y='MonthlyRate', data=avg_salary_gender, palette='coolwarm')
```

```
/tmp/ipykernel_1657705943.py:23: FutureWarning:

Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'x' variable to 'hue' and set 'legend=False' for the same effect.

  sns.barplot(x='MaritalStatus', y='MonthlyRate', data=avg_salary_marital, palette='viridis')
Average Monthly Rate by Marital Status:
  MaritalStatus  MonthlyRate
0      Divorced  14316.801417
1       Married  14844.582129
2        Single  14469.017966

Average Monthly Rate by Education Field:
  EducationField  MonthlyRate
0  Human Resources  24635.000000
1    Life Sciences  14538.152012
2        Marketing  14076.941096
3          Medical  14268.000000
4            Other  13276.700000
5 Technical Degree  14238.363636

/tmp/ipykernel_1657705943.py:36: FutureWarning:

Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'x' variable to 'hue' and set 'legend=False' for the same effect.

  sns.barplot(x='EducationField', y='MonthlyRate', data=avg_salary_education_field, palette='plasma')
```

Since Kobayashi-san will be reviewing your code, please present the data in a way that is clear and accurate. Otherwise, Kobayashi-san might fire you from the company!


```
# Write your code here
# Final Analysis and Summary

# Combine all analyses into a single summary and display it in a single table format
print("\nSummary of Analysis:")

# Gender Analysis
print("\nAverage Monthly Rate by Gender:")
print(avg_salary_gender.to_string(index=False))

# Marital Status Analysis
print("\nAverage Monthly Rate by Marital Status:")
print(avg_salary_marital.to_string(index=False))

# Education Field Analysis
print("\nAverage Monthly Rate by Education Field:")
print(avg_salary_education_field.to_string(index=False))

Summary of Analysis:

Average Monthly Rate by Gender:
Gender  MonthlyRate
Female 14676.608340
  Male 14872.185442

Average Monthly Rate by Marital Status:
MaritalStatus  MonthlyRate
     Divorced 14316.801417
      Married 14844.582129
       Single 14469.017966

Average Monthly Rate by Education Field:
  EducationField  MonthlyRate
Human Resources 24635.000000
  Life Sciences 14538.152012
      Marketing 14076.941096
        Medical 14268.000000
          Other 13276.700000
Technical Degree 14238.363636
```